Akoizo Original Practice - Form 3 - Answer key

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Backbone groups

B. Only sulfur atoms

C. Only DNA bases

D. Only neighboring side-chain colors

Correct: A

2. (2 points) HA contributes to…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Attachment and fusion

B. Only genome replication

C. Only salt sensing

D. Only ATP storage

Correct: A

3. (2 points) An unsupported missing loop should be…

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

A. Presented as directly observed

B. Deleted from the sequence silently

C. Called an active site

D. Marked as missing or modeled

Correct: D

4. (2 points) A disulfide links…

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Any two glycines

B. DNA bases

C. Two phosphate groups

D. Cysteines

Correct: D

5. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Measured free energy

B. Every atom's charge

C. A diagnosis

D. A structural classification

Correct: D

6. (2 points) A protein may require…

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

A. No folding

B. No chemical environment

C. Partners or modifications

D. Only its name

Correct: C

7. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Correct: C

8. (2 points) Model review should use…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Only appearance

B. An unrelated year automatically

C. The verified applicable rubric

D. Invented point values

Correct: C

9. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: A motif is a recurring arrangement; a domain is a larger structural/functional unit. (4 points).

10. (4 points) Why verify residue numbering?

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

answer: Precursor processing, subtype alignments, and deposited chains can use different references. (4 points).

11. (4 points) What should an interaction annotation include?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

answer: Residue identity, geometry, chemical rationale, source, and limitations. (4 points).

12. (4 points) Contrast rate and stability.

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

answer: Rate concerns barriers and pathways; stability concerns relative free energies. (4 points).

13. (4 points) Why is projected overlap insufficient?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

answer: Depth is lost in projection; atoms can appear close while spatially separated. (4 points).

14. (4 points) Why can mature numbering differ?

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

answer: Signal peptides or other segments may be cleaved and annotations can use different references. (4 points).

15. (4 points) What can mutation change?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

answer: Binding, stability, processing, or recognition depending on context. (4 points).

16. (4 points) Why use text with color?

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

answer: It makes the legend accessible and avoids ambiguous color-only meanings. (4 points).

17. (4 points) What does a cartoon hide?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: Many atom-level details and side-chain contacts. (4 points).

18. (4 points) What does a prefusion structure omit?

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

answer: Other conformational states and the full dynamic entry pathway. (4 points).